

Rate UKMA

Deployment infrastructure

UKMA Data on Kubernetes

Two services, two setups

Rate UKMA

- Docker Compose on a VM
- live and staging
- nginx and Certbot

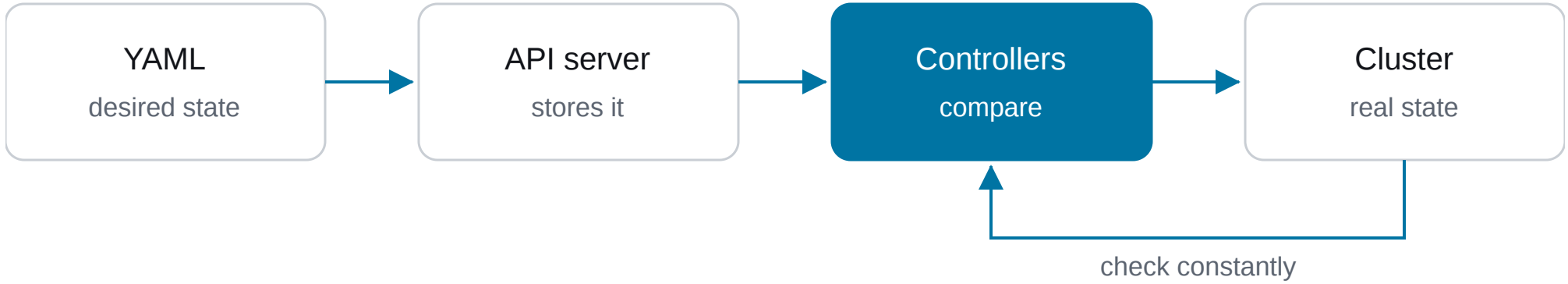
UKMA Data

- Kubernetes on a VM
- staging only for now
- Traefik and operators

The rest of this talk is about UKMA Data only.

The main idea of Kubernetes

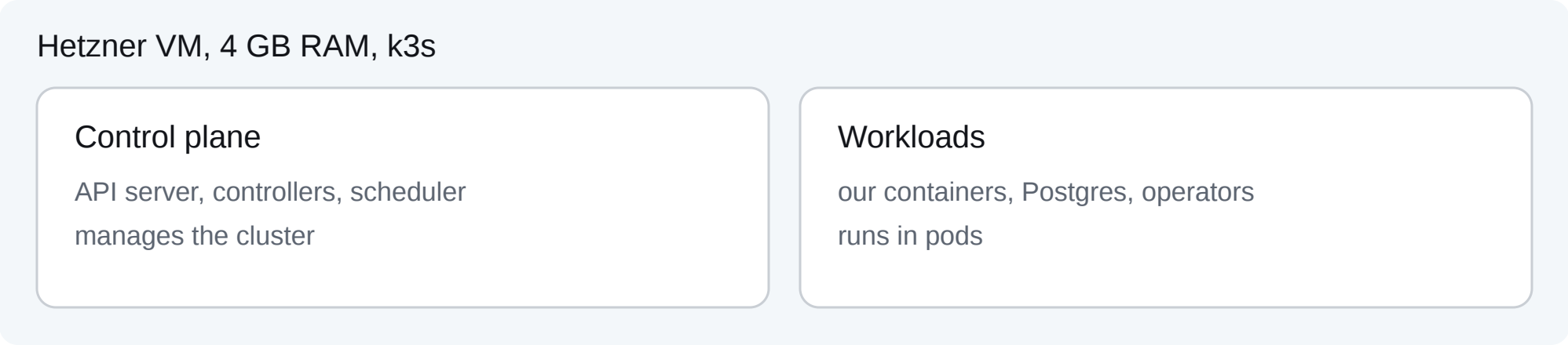
We describe the desired state. Kubernetes gets the cluster there.



Our configs contain no "start" or "restart" commands.

Node and cluster

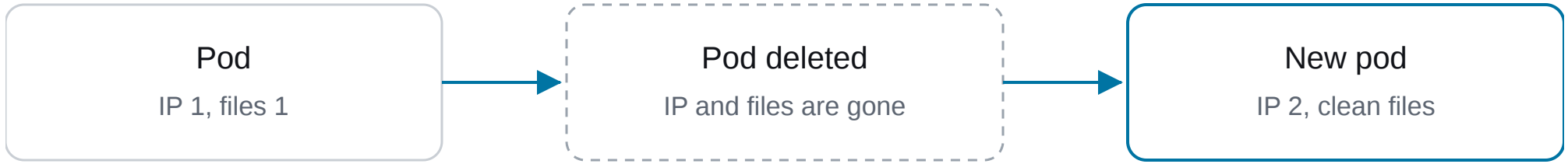
A node is a machine. A cluster is nodes plus a control plane.



Everything we run is on one node. k3s is a lightweight Kubernetes distribution.

Pod

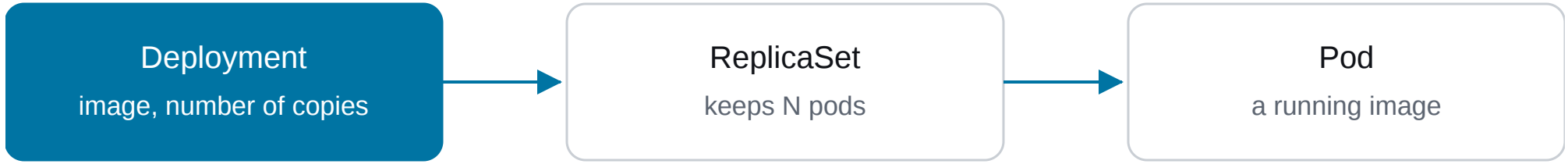
The smallest unit you can run: a container with its own IP.



A pod is temporary. You can't rely on any single pod.

Deployment and ReplicaSet

We don't create pods by hand. We write a Deployment.



Pod gone: the ReplicaSet creates a new one. We never write ReplicaSets.

Updating a Deployment

The old pod keeps running until the new one is ready.



A readiness probe checks this: `GET /health` every 5 s.

Our workloads

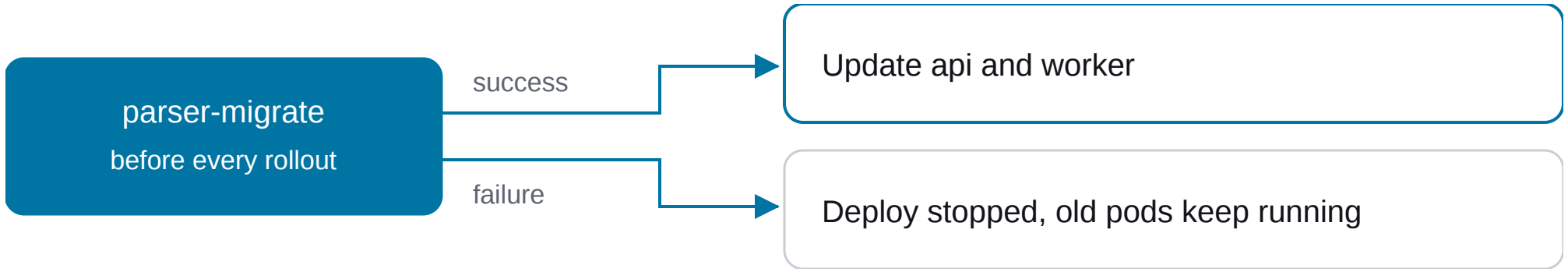
One image, three commands.

Object	Type	Job	Update
`parser-api`	Deployment	HTTP API on port 8080	rolling
`parser-worker`	Deployment	collects SAZ data	Recreate
`parser-migrate`	Job	database migrations	once per deploy

Recreate stops the old pod first, so two workers never run together.

Job

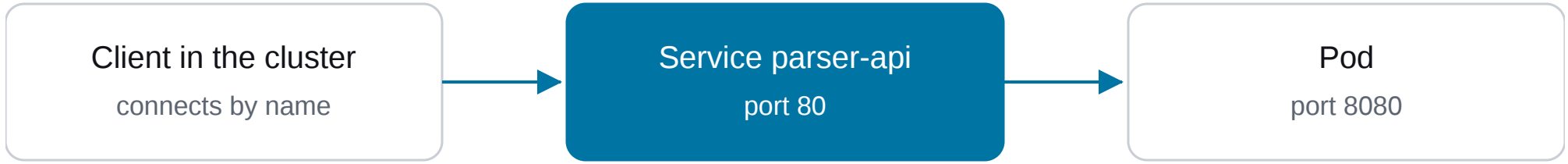
A pod meant to finish. A Deployment would restart it forever.



One attempt, 5 min limit.

Service

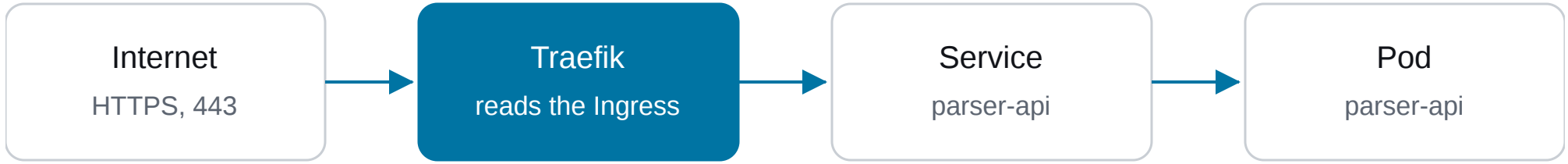
A pod's IP changes. A Service's address stays the same.



A Service finds pods by label, not by IP. It's visible only inside the cluster.

Ingress

A rule: which domain leads to which Service.



An Ingress is only a record. Traefik, built into k3s, carries it out.

Secret

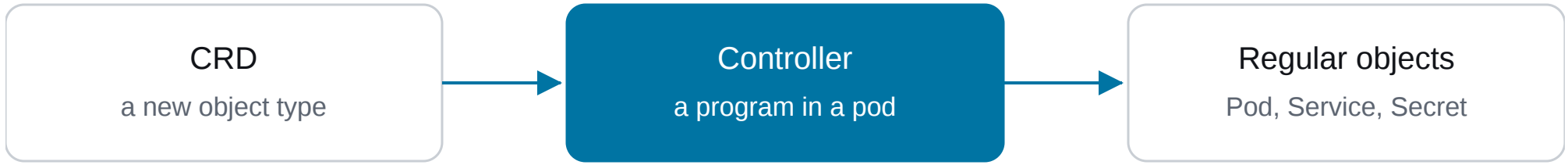
Passwords and keys are kept apart from the YAML.

Secret	Created by	Contains
`parser-db-app`	CloudNativePG	`DATABASE_URL`
`parser-saz`	Parser Platform	SAZ account
`parser-ghcr-pull`	Parser Platform	image access
`parser-api-tls`	cert-manager	TLS certificate

k3s encrypts Secrets on disk.

Operator

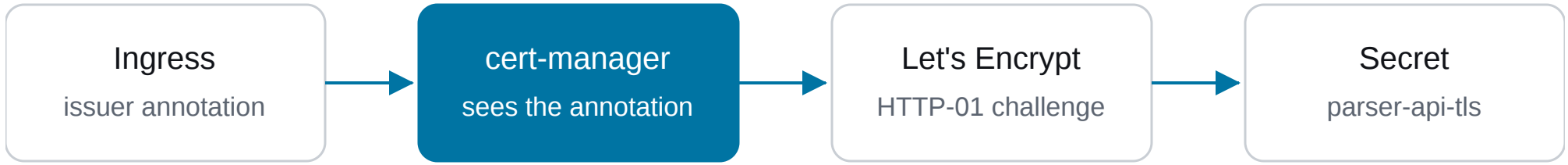
Kubernetes knows Pod and Service, but not Postgres. An operator adds that knowledge.



The same desired-state loop. We have two operators: cert-manager and CloudNativePG.

cert-manager

Obtains and renews TLS certificates.



We added one annotation. No Certbot, no cron.

CloudNativePG

We describe Postgres as one object.

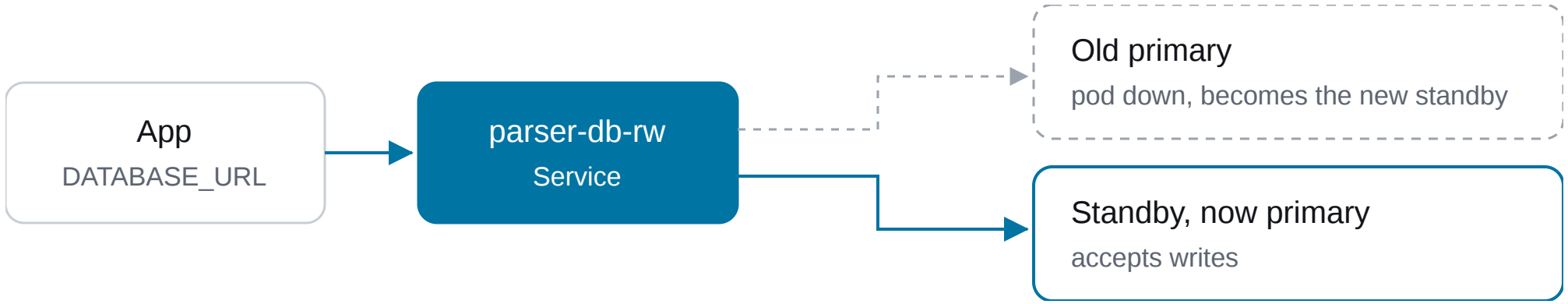
```
kind: Cluster
metadata:
  name: parser-db
spec:
  instances: 2
  storage:
    size: 5Gi
```

The operator creates

- 2 Postgres pods
- replication between them
- Service `parser-db-rw`
- Secret `parser-db-app`

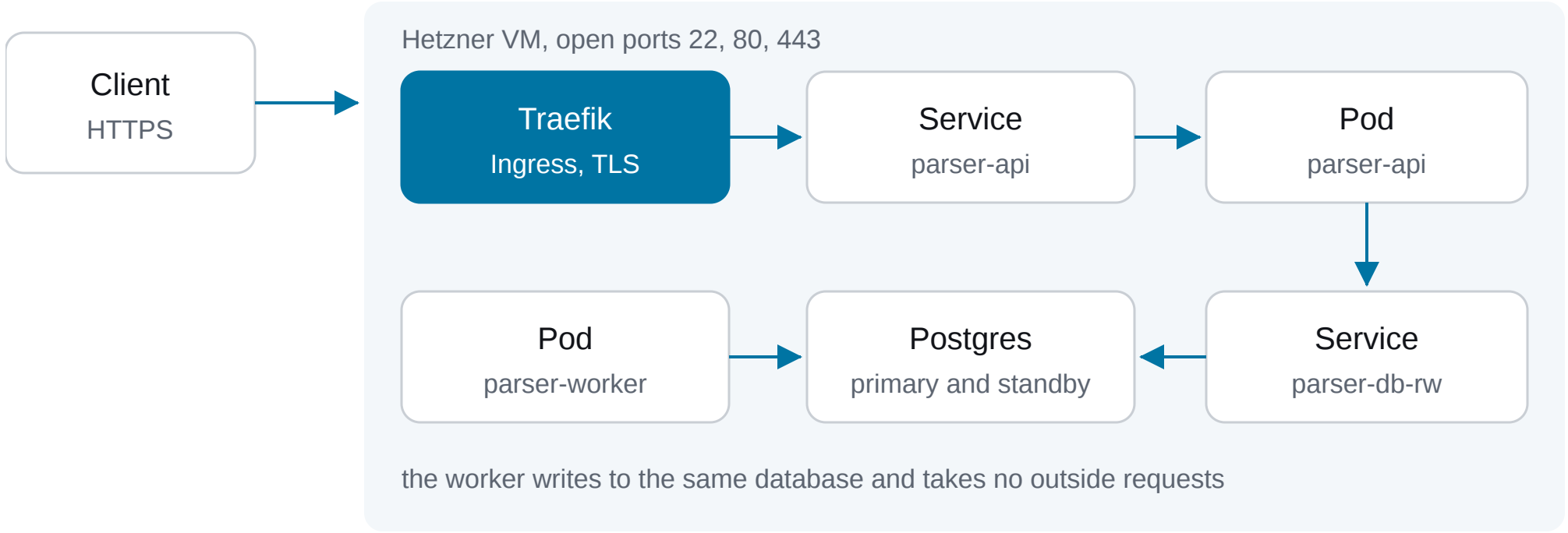
Failover

Primary down: the operator makes the standby the new primary.



The database address doesn't change. The latest commits can be lost.

The full request path



Three layers, three workflows

Each layer is updated separately, through GitHub Actions.

Layer	What it changes	Workflow	Approval
Machine	VM, firewall, k3s	Provision Parser VM	required
Platform	operators, database, Secrets	Parser Platform	required
Application	api, worker, migrations	Deploy Parser	not required

The lower the layer, the less often it changes and the more a mistake costs.

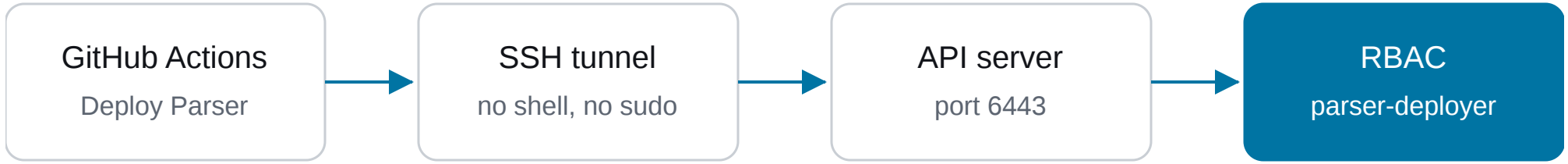
Deploying the application



Started by hand: a version and an environment. One deploy per environment at a time.

Cluster access

The Kubernetes API is closed to the internet.



Only the `parser` namespace. It can't read Secrets or change permissions.

Current limits

Reliability

- one node: VM down, everything down
- no database backups yet
- data only on the VM's disk

Not done yet

- no live, staging only
- deploys start by hand
- no cluster monitoring

Glossary

Term	What it is
Node	a machine in the cluster
Pod	a running container, temporary
ReplicaSet	keeps the right number of pods
Deployment	an app version, updates its pods
Job	a pod that's meant to finish
Service	a permanent address for pods
Ingress	a rule for incoming HTTP traffic
Secret	passwords and keys for pods
Namespace	a named group of objects
Operator	a CRD plus a controller